

Public Sentiment and Common Topics on Philippine Agriculture: A Comparative Text Analysis of Reddit Discussions and Online News Articles

Louis G. Binwag III

Department of Information Systems and Computer Science

Ateneo De Manila University

Quezon City, Philippines

louis.binwag@student.ateneo.edu

Abstract—This study focuses on public sentiments and common topics of agriculture in the Philippines, specially in news platforms and reddit.

Index Terms—component, formatting, style, styling, insert

I. INTRODUCTION

A. Background and Brief Related Literature

The main intuition in this study is to provide context into what are the common topics and sentiments towards the topic **agriculture** in the Philippines. While it is common knowledge that agriculture is often **unappreciated** and that it is **unsupported**, this study provides a textual analysis on the conversations that occur in online platforms.

The Department of Agriculture (DA) recognizes the fact that the agricultural situation in the Philippines is struggling [1]. These struggles are stem from various challenges that the sector is facing, such as the decreasing number of farmers, harvestable area, and the increasing cost of production [2]. With the continuous rise of population, the demand for food is also increasing. However, the agricultural sector is not able to keep up with this demand due to various challenges such as lacking of government support, climate change, and limited access to new farming technology [2].

Considering the challenges that the agricultural sector is facing, it is important to understand the public sentiments and common topics associated with agriculture in the Philippines. This study aims to analyze the public sentiments and common topics associated with agriculture in the Philippines by utilizing text data from Reddit discussions and rappler news articles. By comparing the sentiments and topics from these two, we will be able to identify the differences and similarities in the public discourse on agriculture in the Philippines.

B. Research Questions

- 1) What are the dominant public sentiments toward agriculture in the Philippines across Reddit discussions and news articles?

- 2) What are the most common topics associated with Philippine agriculture in public (Reddit) versus formal (news) discourse?
- 3) How do public sentiments and topics differ between the two sources?
- 4) Is it possible to identify the specific needs or concerns of farmers based on these online discussions?

C. Scope & Limitations

- 1) Considering the topic utilizes a discussion-based dataset, the study will not take into account the network structure of the data.
- 2) The analysis is limited to text content and does not incorporate multimedia elements such as images or videos, which may also influence discussions or public perception.

II. METHODOLOGY

TABLE I
RAW REDDIT DATA SAMPLE

post_title	Sample title about farming
post_url	https://reddit.com/...
comment	Agriculture in the Philippines needs reform.
comment_author	user123
comment_score	12
post_body	Full post discussing rice production and its challenges in the Philippines.
post_score	256
post_author	authorA
created_utc	2025-10-23 08:14:00

TABLE II
ONLINE NEWS PLATFORM DATA SAMPLE

title	Arms to farms: Maguindanao del...
link	https://www.rappler.com/philippines/...
date_published	2025-10-24 09:30:00
author	Maria Dela Cruz
text	COTABATO CITY, Philippines —Maguindanao del...

A. Data Collection & Preprocessing

Reddit data is collected from r/Philippines, while agriculture-related news articles are taken from **Rappler**. Metadata that are gathered from both sources are summarized in their respective tables (*See Table I & II*) The data is cleaned by (i) removing null values, (ii) removing special characters, (iii) normalizing text, and (iv) identifying and removing stopwords.

Exploratory Data Analysis

At this stage, basic exploratory data analysis (EDA) methods are going to provide us a deeper understanding of what the data has to offer. (i) Basic Overview and Structure, (ii) Word Frequency Analysis, (iii) Engagement and Metadata Trends, (iv) Common Keywords, (v) Text Comparison (*Reddit vs News, how do they write text?*)

Natural Language Processing a. Topic Modelling *b. Sentiment Analysis*

III. RESULTS & DISCUSSION

IV. CONCLUSION

V. POSSIBLE FUTURE WORKS

REFERENCES

REFERENCES

- [1] DA Press Office, "Facing the Big Challenges in Philippine Agriculture," 2022. [Online]. Available: <https://www.da.gov.ph/facing-the-big-challenges-in-philippine-agriculture/>. [Accessed: 9- Nov- 2025].
- [2] GMA News, "Bakit Hindi Rice Self-Sufficient Ang Pilipinas? — Need to Know," 2021. [Online]. Available: <https://www.youtube.com/watch?v=nh3vLr9DG1g>. [Accessed: 20-Oct- 2025].

IEEE conference templates contain guidance text for composing and formatting conference papers. Please ensure that all template text is removed from your conference paper prior to submission to the conference. Failure to remove the template text from your paper may result in your paper not being published.